Question:

A 34-year-old poultry worker presents to his physician with a sore throat and a non-productive cough for 2 weeks. His cough is associated with fever. The vital signs include: blood pressure 120/80 mm Hg, heart rate 67/min, respiratory rate 18/min, and temperature 37.6°C (98.0°F). Physical examination shows oropharyngeal erythema and scattered, moist rales on lung auscultation. The patient’s X-ray demonstrates patchy reticular opacities in the perihilar regions of both lungs. After some additional tests, he is diagnosed with community-acquired pneumonia and is treated with cephalexin with modest improvement. Which of the following best describes the immune response elicited by the pathogen that is causing this patient’s condition?

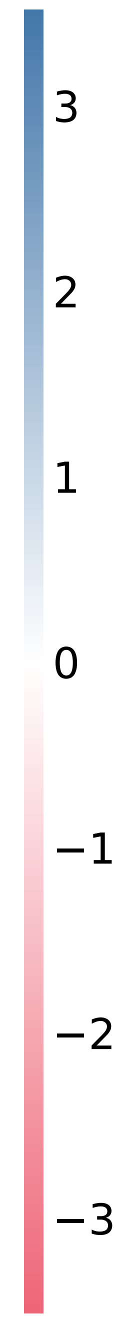
Answer Choices:

A. It activates TLR5 on the surface of macrophages.

B. It primarily induces the Th1-cell response.

C. Peptidoglycan is its major antigen that induces an immune response.

D. This pathogen evades the immune response by encapsulation.



Reasoning + Answer:

<think>Given that the patient exhibits symptoms like sore throat and cough, indicative often of viral infections, and given the pulmonary imaging showing reticulations, suggesting presence of bacterial invasion, hereby suggesting a community-acquired bacterial pneumonia, leading to infer that the agent could be Streptococcus pneumoniae, a pathogen known to encapsulate itself to evade immune response, thereby initiating defense through such mechanism.</think><answer> This pathogen evades the immune response by encapsulation.</answer>